André **Anjos**

HEAD OF THE MEDICAL ARTIFICIAL INTELLIGENCE GROUP, PH.D.



About me

André Anjos leads the Medical AI Group at the Idiap Research Institute, working on machine learning, computer vision, and signal processing for medical decision making, research reproducibility, and open science.

Interests

- Pattern Recognition
- Machine Learning
- Reproducible Research
- Computer Vision
- Signal Processing
- Artificial Intelligence
- Medical AI

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Personal

Date of birth: 31.12.1973 (52 years old)
Nationality: Swiss and Brazilian
Family: Married, one child

anjos.ai

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medai

PROFESSIONAL EXPERIENCE

2018 – present **Scientific Researcher**

Idiap Research Institute, Martigny, Switzerland
Head of the Medical Artificial Intelligence Group.

2019 – 2023 **Lecturer**

Idiap Research Institute & Unidistance Suisse, Martigny, Switzerland
Three modules of the Master in Artificial Intelligence: open science and ethics (M05), and fundamentals of machine learning (M06 and M08).

2013 – 2023 **Lecturer**

EPFL (École Polytechnique Fédérale de Lausanne), Lausanne, Switzerland
EE-612, Fundamentals of Statistical Pattern Recognition: a doctoral course on machine learning and pattern recognition, from the basics to deep learning, given biennially.

2014 – 2018 **Research Associate**

Idiap Research Institute, Martigny, Switzerland
Research on biometrics, security and computing.

2010 – 2013 **Post-doctoral Researcher**

Idiap Research Institute, Martigny, Switzerland
Research on biometrics, security and computing.

2004 – 2010 **Research & Development Engineer**

CERN, Switzerland

University of Wisconsin, Madison, USA
Design, development and maintenance of the ATLAS Trigger and Data Acquisition systems, deployed and operational on more than 5000 machines at CERN and worldwide.

2000 – 2004 **Research Assistant**

CERN, Switzerland

UFRJ (Federal University of Rio de Janeiro), Brazil
Design, implementation and integration of a novel algorithm for fast physics triggering — based on calorimetry, topological mapping and neural networks — into the second-level trigger software of the ATLAS Experiment, alongside the online infrastructure needed to run and validate it.

1997 – 1998 **Research Internship**

CERN, Switzerland
Contributed to early prototypes of the second-level trigger and to what became the final trigger design of the ATLAS Experiment.

1994 – 1999 **Research Internship**

UFRJ (Federal University of Rio de Janeiro), Rio de Janeiro, Brazil
Sub-optimal filtering for particle discrimination, and neural classifiers implemented in a transputer-based parallel machine.

EDUCATION

2001 – 2006 **Ph.D., Electronics Engineering (Signal Processing & Machine Learning)**

UFRJ (Federal University of Rio de Janeiro), Rio de Janeiro, Brazil
Electronics Engineering School (Signal Processing Laboratory), Post-Graduate Program (COPPE).
Thesis developed at CERN, Switzerland, in the context of the ATLAS Experiment.

2000 – 2001 **M.Sc., Electronics Engineering (Signal Processing & Machine Learning)**

UFRJ (Federal University of Rio de Janeiro), Rio de Janeiro, Brazil
Electronics Engineering School (Signal Processing Laboratory), Post-Graduate Program (COPPE).
Thesis developed in the context of the ATLAS-CERN / UFRJ collaboration.

1994 – 1999 **Engineer, Electronics Engineering**

UFRJ (Federal University of Rio de Janeiro), Rio de Janeiro, Brazil
Degree from the Electronics Engineering Department.

1989 – 1993 **Technician, Electronics**

CEFET (Federal Center of Technology), Rio de Janeiro, Brazil
Technical course in electronics, taken alongside high-school studies.

Languages

Portuguese — Mother tongue

English — Fluent

French — Fluent, certified C1

Spanish — Basic, reads well

Computer Skills

- Linux, macOS, Windows
- System and network administration
- Structured and object-oriented languages
- Concurrent and parallel programming
- Large software projects, releases and policies
- Revision control, CI, documentation
- Database design and use

Research Areas

2023–2026 **Ophthalmology — Uveitis**

Hôpital ophtalmique Jules-Gonin (HOJG), Lausanne; University Hospital of Grenoble Alpes; Luzerner Kantonsspital

AI-assisted grading of intraocular inflammation in uveitis from retinal fluorescein angiography, developed with the Jules-Gonin Eye Hospital.

anjos.ai/projects/uveitis

2017 **Neurophysiology — Sleep Medicine**

Generalisable, reproducible automatic sleep-stage classification from polysomnography, working toward foundation models for sleep medicine.

anjos.ai/projects/sleep-medicine

2023–2026 **Trustworthy and Fair Medical AI**

Federal University of São Paulo (UNIFESP)

Making medical AI fair across patient subgroups and useful in the clinic, with evaluation frameworks, foundation-model bias analysis, and utility-fairness trade-offs.

anjos.ai/projects/trustworthy-fair-ai

2020–2024 **Ophthalmology — Retinal Image Analysis**

University Hospital of Grenoble Alpes; Idiap Research Institute

Segmentation and analysis of retinal fundus images (vessels, optic disc, blood flow) with minimalistic, reproducible, and clinically deployable models.

anjos.ai/projects/retinal-image-analysis

2021–2024 **Radiology — Robust Image Analysis**

Federal University of Rio de Janeiro; University of Zürich; Murabei Data Science, Brazil; University Hospital Basel; Lausanne University Hospital (CHUV); University of Geneva; ETH Zurich

Interpretable, generalisable computer-aided analysis of radiological images — detecting tuberculosis on chest X-rays and extracting stable quantitative biomarkers from CT — built to hold up across scanners and populations.

anjos.ai/projects/tuberculosis-radiology

2020–2025 **Epidemiology — Anticipating Risk**

University of Geneva; Polytechnic University of Leiria; Federal University of Rio de Janeiro; State University of Campinas (UNICAMP); Federal University of Lavras

Machine learning on routine clinical records, chest radiographs, and patient-contact networks to flag who is most at risk — from diabetic-foot complications to drug-resistant infection to tuberculosis.

anjos.ai/projects/tuberculosis-epidemiology

Angiology — Fibromuscular Dysplasia

CHUV, Lausanne University Hospital

Machine learning for the characterisation and diagnosis of fibromuscular dysplasia, in collaboration with CHUV's angiology department.

anjos.ai/projects/fibromuscular-dysplasia

2015–2026 **Reproducible Research and Open Science**

Software suites and open platforms that make computational research repeatable, shareable, extensible, and stable, including the BEAT open-science platform.

anjos.ai/projects/reproducibility

2016–2017 **Remote Photoplethysmography and Vital Signs**

Reproducible, camera-based estimation of vital signs, putting remote photoplethysmography on a shared, openly benchmarked footing.

anjos.ai/projects/rppg-vital-signs

2011–2019 **Biometrics and Presentation-Attack Detection**

Biometrics Security & Privacy Group, Idiap

Earlier work with the Idiap Biometrics group on robust recognition and presentation-attack detection that generalises to unseen attacks.

anjos.ai/projects/biometrics-pad

1998–2009 **High-Energy Physics — ATLAS Trigger (CERN)**

CERN, ATLAS Experiment; Federal University of Rio de Janeiro

Earlier work at CERN's ATLAS experiment on real-time trigger and data-acquisition systems, and neural-network methods for particle recognition.

anjos.ai/projects/hep-cern

Grants and Funding

2026 – 2030 **IMAGIN-AIR: From Images to Knowledge: AI to Elucidate Mechanisms and Improve Therapeutic Strategies for Retinal Inflammation**

Carigest

www.idiap.ch/en/projects/imagin-air

2025 – 2028 **AI2Pub: AI Insights to the Public through bi-directional media channels**

Swiss National Science Foundation (Instrument: Agora)

data.snf.ch/grants/grant/232373

2024 – 2028 **FairMI: Machine Learning Fairness with Application to Medical Images**
Swiss National Science Foundation (Instrument: Lead Agency)
data.snf.ch/grants/grant/214653

2025 – 2027 **DREAM: Detection and Research for Learning Sets on Electroencephalographic Patterns**
Assistance Publique – Hôpitaux de Paris
www.idiap.ch/en/projects/dream

2025 – 2026 **VALIDATE-H: VALaIs aDaptation of Artificial inTelligence modEls in Healthcare**
Service des Hautes Écoles
www.idiap.ch/en/projects/validate-h

2024 – 2025 **CollabCloud: A Collaborative Research Cloud Infrastructure powering Discovery and Exchange**
Swiss National Science Foundation
data.snf.ch/grants/grant/220409

2023 – 2024 **NewsOnAI: Scientific voice to latest news of AI technology**
Swiss National Science Foundation
data.snf.ch/grants/grant/214101

2023 – 2023 **Nouvelle plateforme de recherche « Cloud » pour étendre les capacités de recherche, d'échange et de formation de l'Idiap**
Loterie Romande
loro.ch/fr/beneficiaires/projets-soutenus/collabcloud

2022 – 2023 **CAD4IED: Computer Aided Diagnosis for Inflammatory Eye Diseases**
Idiap Research Institute
www.idiap.ch/en/projects/cad4ied

2019 – 2023 **LEARN-REAL: LEARNing physical manipulation skills with simulators using REAListic variations**
Swiss National Science Foundation (Instrument: CHIST-ERA)
data.snf.ch/grants/grant/180417

2022 – 2023 **SECure: Safe and Explainable Clinical AI for Orthopaedic Surgical Assessment**
TheArk

2019 – 2020 **SEWS2: Smart Early Warning Score System for in- and out-hospital care via anomaly detection**
Innosuisse – Swiss Innovation Agency (Instrument: Innovation check)

Earlier Career Contributions

Mostly while working at the Biometrics & Security group of Sébastien Marcel.

2019 – 2021 **AI4EU: A European AI On Demand Platform and Ecosystem**
European Commission (Instrument: Horizon 2020)
Roles: Co-authorship of the Idiap contribution, project management
www.idiap.ch/en/projects/ai4eu

2018 – 2021 **ALLIES: Autonomous Lifelong learnIng intelLigent Systems**
Swiss National Science Foundation (Instrument: CHIST-ERA)
Roles: Co-authorship, project management, implementation
data.snf.ch/grants/grant/174239

2017 – 2021 **BATL: Biometric Authentication with a Timeless Learner**
US Intelligence Advanced Research Projects Activity (IARPA) (Instrument: Odin program)
Roles: Co-authorship of the Idiap participation (face modality)
www.idiap.ch/~marcel/project/former/batl

2012 – 2016 **BEAT: Biometrics Evaluation and Testing**
European Commission (Instrument: Seventh Framework Programme (FP7))
Roles: Co-authorship, implementation, project management
www.idiap.ch/en/projects/beat

2014 – 2015 **COHFACE: COntactless Heartbeat detection for trustworthy FACE Biometrics**
Hasler Foundation
Roles: Co-authorship, dataset acquisition management, implementation
www.idiap.ch/en/projects/cohface

2013 – 2014 **FEDARS: Feature Extraction from Deep learning Architectures for face Recognition Systems**
Swiss Commission for Technology and Innovation (CTI)
Roles: Implementation
www.idiap.ch/en/projects/fedars

2010 – 2014 **TABULA RASA: Trusted Biometrics under Spoofing Attacks**
European Commission (Instrument: Seventh Framework Programme (FP7))

Roles: Dataset acquisition (WP2), implementation (WP4)
www.idiap.ch/en/projects/tabularasa

Teaching

2019–2023 **Master in Artificial Intelligence: Modules M05, M06 and M08**
Idiap & Unidistance
Three modules taught between 2019 and 2023 to master students of the Master in AI run by the Idiap Research Institute with Unidistance: open science and ethics (M05), and machine learning fundamentals (M06 and M08).

2013–2023 **Fundamentals of Statistical Pattern Recognition**
EPFL, Lausanne
EE-612 at EPFL: fundamental tools of machine learning and pattern recognition, from basics to deep learning, for Ph.D. students. Given biennially from 2013 to 2023.

2022 **IFI/UZH Summer School: From Scripts to Reusable Software and Reproducible Research**
University of Zurich
Summer school at the University of Zurich: actionable guidelines for turning non-reproducible scripts with hard-coded parameters into reusable software, with examples in Python.

2015 **Reproducible Research for Pattern Recognition**
UNESP Bauru, Brazil
A hands-on course given at UNESP Bauru, Brazil, on reproducible research for engineers working with software in pattern recognition and machine learning.

Supervision

2026 **Alexis Bollengier**
Master thesis — VUB, Brussels Faculty of Engineering (BRUFACE)
Sleep Spindle Detection from Electroencephalography via Sleep Foundation Models
www.idiap.ch/~aanjos/pdfs/theses/alexis-bollengier.pdf

2026 **Tymo van Rijn**
Bachelor thesis — Hogeschool Rotterdam
Beyond the Last Frame: Temporal Deep Learning for Automated Grading of Retinal Inflammation in Fluorescein Angiography
www.idiap.ch/~aanjos/pdfs/theses/tymo-van-rijn.pdf

2025 **Daniel Lutziger**
Master thesis — University of Zürich, Department of Informatics
Towards Scalable Foundation Models for Sleep EEG and Polysomnography
seafile.ifi.uzh.ch/f/9951eb6cc9c64b0fa330

2025 **Maximilian Achakri**
Master thesis — University of Zürich, Department of Informatics
Towards Computer-Aided Diagnosis of Fibromuscular Dysplasia: Abnormality Detection in the Renal Artery Using Deep Learning
www.idiap.ch/~aanjos/pdfs/theses/maximilian-achakri.pdf

2025 **Reto Odoni**
Master thesis — University of Zürich, Department of Informatics
Measuring Bias in AI Anatomical Structure Segmentation Models
seafile.ifi.uzh.ch/f/da14adc2a0a14e70964d

2025 **Roberto Pulvirenti**
Master thesis — Politecnico di Torino, MSc in Artificial Intelligence & Data Analytics
Evaluating Cross-Domain Adaptation Strategies for Foundation Models: A Study on Fluorescein Angiography
www.idiap.ch/~aanjos/pdfs/theses/roberto-pulvirenti.pdf

2024 **Joël Inglin**
Master thesis — University of Zürich, Department of Informatics
Stability and Accuracy of Radiomic Features in Medical Image Classification
seafile.ifi.uzh.ch/f/4a5dbaf45523404d860d

2024 **Yvan Pannatier**
Master thesis — University of Geneva, Faculty of Science (MSc Computer Science)
Clinically Interpretable Computer-Aided Diagnostic Tool for Fibromuscular Dysplasia Detection
www.idiap.ch/~aanjos/pdfs/theses/yvan-pannatier.pdf

2023 **Gabriele Brunini**
Master thesis — University of Zürich, Department of Informatics
Automatic Sleep-Phase Analysis via Stateful Methods

2023 **Geoffrey Raposo**
Master thesis — UniDistance, Brig (Master in AI, jointly with Idiap)

Active Tuberculosis Detection from Frontal Chest X-ray Images
http://publications.idiap.ch/attachments/reports/2021/Raposo_Idiap-Com-01-2021.pdf

2023 **Özgür Güler**

Master thesis — University of Zürich
Explaining CNN-Based Active Tuberculosis Detection through Saliency Mapping
capuana.ifi.uzh.ch/publications/PDFs/24103_Master_Thesis_oezguer_acar_gueler.pdf

2023 **Samuel Michel**

Master thesis — UniDistance, Brig (Master in AI, jointly with Idiap)
Generalizable Automatic Classification of Sleep Stages
publications.idiap.ch/attachments/reports/2023/Michel_Idiap-Com-02-2023.pdf

2023 **Victor Amiot**

Master thesis — EPFL, School of Life Sciences
Automatic Grading of Vasculitis Inflammation in Fluorescein Angiography Images
www.idiap.ch/~aanjos/pdfs/theses/victor-amiot-vasculitis.pdf

2023 **Xiao Tan**

Master thesis — University of Zürich, Department of Informatics
Semantic Segmentation of Weakly Labeled Retinal Images
capuana.ifi.uzh.ch/publications/PDFs/23611_Xiao%20Tan_thesis.pdf

2022 **Antonio Morais**

Master thesis — EPFL, Electrical Engineering
Is System A Statistically Better Than System B?
www.idiap.ch/~aanjos/pdfs/theses/antonio-morais.pdf

2022 **Driss Khalil**

Master thesis — UniDistance, Brig (Master in AI, jointly with Idiap)
Computer-Aided Screening for Eye Diseases from 2-D Retinography
www.idiap.ch/~aanjos/pdfs/theses/driss-khalil.pdf

2022 **Maxime Délitroz**

Master thesis — EPFL
Automated Segmentation of Developing Motor Neurons in Fluorescence Microscopy

2020 **Colombine Verzat**

Master thesis — UniDistance, Brig (Master in AI, jointly with Idiap)
Machine Learning for Adverse-Event Detection in Latent Tuberculosis Treatment
publications.idiap.ch/attachments/reports/2020/Verzat_Idiap-Com-02-2020.pdf

2015 **Ivana Chingovska**

De facto co-supervisor — EPFL, Electrical Engineering (doctoral programme, with Idiap)
Trustworthy Biometric Verification under Spoofing Attacks: Application to the Face Mode
publications.idiap.ch/attachments/papers/2016/Chingovska_THESIS_2015.pdf

2018 **Jaden Diefenbaugh**

Graduate student — Idiap Research Institute, Switzerland
Work associated with the BEAT platform.

2010 – 2013 **Tiago de Freitas Pereira and Jukka Määttä**

Visiting students, co-supervision — Idiap Research Institute, Switzerland
Presentation-attack detection in biometrics.

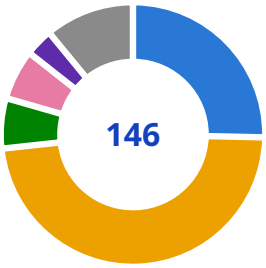
2010 – 2013 **Murali Mohan Chakka**

Trainee, co-supervision — Idiap Research Institute, Switzerland
Presentation-attack detection in biometrics.

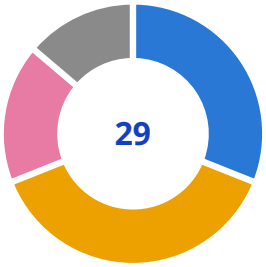
Bibliometrics

h-index: 32
Citations: 29'000+
Google Scholar, 2026

Research Outputs



All-time



Last 5y

	all	5y
Journals	37	9
Conferences	70	11
Chapters	9	0
Software	9	5
Datasets	5	0
Other	16	4

Other: Preprint, Report, Patent, Thesis, Press, Presentation.

Open-Source Software

2025 **fairical**
A Python library for evaluating and tuning machine-learning systems via multi-objective optimisation, balancing fairness metrics against utility; it approximates Pareto fronts of utility–fairness trade-offs with convergence, diversity, and hypervolume scores.
gitlab.idiap.ch/medai/software/fairical

2024 **mednet**
A framework for developing and analysing deep neural network architectures for medical data (2D/3D images and signals), with ready access to many public medical-imaging datasets and extensible model architectures.
gitlab.idiap.ch/medai/software/mednet

2023 **clapper**
Configuration and command-line-interface support for Python packages: automatic TOML-based RC configuration storage/retrieval and integration with the Click framework for building CLIs.
github.com/idiap/clapper

2023 **credible**
scikit-learn-compatible implementations of machine-learning metrics augmented with credible/confidence intervals, so performance measurements come with quantified uncertainty bounds.
gitlab.idiap.ch/medai/software/credible

2023 **auto-intersphinx**
A Sphinx extension that fills intersphinx mappings from package names, removing the need to manually specify dependency documentation URLs and keeping cross-references up to date.
github.com/idiap/auto-intersphinx

2017 **sleepless**
A benchmarking package for evaluating machine- and deep-learning techniques on polysomnography biosignals for sleep-stage classification, covering feature extraction, training, prediction, and evaluation across datasets and algorithms.
gitlab.idiap.ch/biosignal/software/sleepless

2016 **BEAT**
A web-based open-science platform for running, sharing, and reusing reproducible machine-learning experiments with secure access to data and code, even when the underlying data cannot be published.
gitlab.idiap.ch/beat

2015 **Bob**
A free signal-processing and machine-learning toolbox developed at Idiap: a comprehensive, Python-first framework to build, experiment with, and reproduce pattern-recognition and machine-learning workflows.
gitlab.idiap.ch/bob/bob

2012 **gridtk**
A user-friendly command-line tool for Slurm job management — a drop-in replacement for sbatch adding local job IDs, automatic log management, monitoring, and simplified job dependencies.
github.com/idiap/gridtk

Open Datasets

2019 **WMCA (Wide Multi-Channel presentation Attack)**
doi.org/10.34777/8zdh-v182

2016 **COHFACE**
doi.org/10.34777/ff3f-ba56

2015 **MSSpoof (Multispectral-Spoof)**
doi.org/10.34777/payf-vb10

2012 **Replay-Attack**
doi.org/10.34777/cwcg-7r82

2011 **Print-Attack**

Journal Articles

2026 V. Amiot, R. Pulvirenti, O. Jimenez-del-Toro, M. Ott, T. Bogaciu, S. Banerjee, C. Amstutz, J. Odobez, C. Chiquet, Y. Guex-Crosier, C. Bergin, I. Meloni, **A. Anjos**, F. Hoogewoud, and M. Tomasoni, "UveAI: clinic-ready scoring of retinal inflammation in uveitis on widefield fluorescein angiography using AI", *Scientific reports*, 2026, doi: *10.1038/s41598-026-46069-w*.

D. Q. Neto, A. Carlos, **A. Anjos**, and L. Berton, "Fair Foundation Models for Medical Image Analysis: Challenges and Perspectives", *ACM Transactions on Computing for Healthcare*, 2026, doi: *10.1145/3793542*.

2025 G. Özbulak, O. Jimenez-del-Toro, M. Fatoretto, L. Berton, and **A. Anjos**, "A Multi-Objective Evaluation Framework for Analyzing Utility-Fairness Trade-Offs in Machine Learning Systems", *The Journal of Machine Learning for Biomedical Imaging 3:938-957*, 2025, doi: *10.59275/j.melba.2025-ab9a*.

M. Fatoreto, G. Özbulak, L. Berton, and **A. Anjos**, "Optimizing Fairness and Utility in Healthcare Machine Learning Models", *Applied Soft Computing 181(113426)*, 2025, doi: *10.1016/j.asoc.2025.113426*.

V. Amiot, O. Jimenez–del–Toro, Y. Guex–Crosier, M. Ott, T. Bogaciu, S. Banerjee, J. Howell, C. Amstutz, C. Chiquet, C. Bergin, I. Meloni, M. Tomasoni, F. Hoogewoud, and **A. Anjos**, "Automatic transformer-based grading of multiple retinal inflammatory signs in uveitis on fluorescein angiography", *Computers in Biology and Medicine 193:110327*, 2025, doi: *10.1016/j.compbiomed.2025.110327*.

2024 T. Mautuit, P. Cunnac, F. Truffer, **A. Anjos**, R. Dufrane, G. Maître, M. Geiser, and C. Chiquet, "Absolute retinal blood flow in healthy eyes and in eyes with retinal vein occlusion", *Microvascular Research 152:104648*, 2024, doi: *10.1016/j.mvr.2023.104648*.

2023 C. Geric, Z. Z. Qin, C. M. Denkinger, S. V. Kik, B. Marais, **A. Anjos**, P. David, F. A. Khan, and A. Trajman, "The rise of artificial intelligence reading of chest X-rays for enhanced TB diagnosis and elimination", *The International Journal of Tuberculosis and Lung Disease*, 2023, doi: *10.5588/ijtld.22.0687*.

2022 M. Shamsi, A. Larcher, L. Barrault, S. Meignier, Y. Prokopalo, M. Tahon, A. Mehrish, S. Petitrenaud, O. Galibert, S. Gaist, **A. Anjos**, S. Marcel, and M. R. Costa-jussà, "Towards lifelong human assisted speaker diarization", *Computer Speech & Language:101437*, 2022, doi: *10.1016/j.csl.2022.101437*.

A. Galdran, **A. Anjos**, J. Dolz, H. Chakor, H. Lombaert, and I. B. Ayed, "State-of-the-art retinal vessel segmentation with minimalistic models", *Scientific Reports 12(1):6174*, 2022, doi: *10.1038/s41598-022-09675-y*.

2020 A. C. B. H. Ferreira, D. D. Ferreira, H. C. Oliveira, I. C. de Resende, **A. Anjos**, and M. H. B. M. Lopes, "Competitive neural layer-based method to identify people with high risk for diabetic foot", *Computers in Biology and Medicine 120:103744*, 2020, doi: *10.1016/j.compbiomed.2020.103744*.

2019 T. F. Pereira, **A. Anjos**, and S. Marcel, "Heterogeneous Face Recognition Using Domain Specific Units", *IEEE Transactions on Information Forensics and Security*, 2019, doi: *10.1109/tifs.2018.2885284*.

A. George, Z. Mostaani, D. Geissenbuhler, O. Nikisins, **A. Anjos**, and S. Marcel, "Biometric Face Presentation Attack Detection with Multi-Channel Convolutional Neural Network", *IEEE Transactions on Information Forensics and Security*, 2019, doi: *10.1109/tifs.2019.2916652*.

2016 A. Morales, J. Fierrez, R. Tolosana, J. Ortega-Garcia, J. Galbally, M. Gomez-Barrero, **A. Anjos**, and S. Marcel, "Keystroke Biometrics Ongoing Competition", *IEEE Access 4:7736–7746*, 2016, doi: *10.1109/access.2016.2626718*.

2015 I. Chingovska, and **A. Anjos**, "On the use of client identity information for face anti-spoofing", *IEEE Transactions on Information Forensics and Security, Special Issue on Biometric Anti-spoofing 10(4):787–796*, 2015, doi: *10.1109/tifs.2015.2400392*.

2014 T. F. Pereira, J. Komulainen, **A. Anjos**, J. M. De Martino, A. Hadid, M. Pietikainen, and S. Marcel, "Face liveness detection using dynamic texture", *EURASIP Journal on Image and Video Processing 2014:2*, 2014, doi: *10.1186/1687-5281-2014-2*.

I. Chingovska, **A. Anjos**, and S. Marcel, "Biometrics Evaluation Under Spoofing Attacks", *IEEE Transactions on Information, Forensics and Security 9(12)*, 2014, doi: *10.1109/tifs.2014.2349158*.

2013 **A. Anjos**, M. M. Chakka, and S. Marcel, "Motion-Based Counter-Measures to Photo Attacks in Face Recognition", *IET Biometrics*, 2013, doi: *10.1049/iet-bmt.2012.0071*.

2009 R. Torres, **A. Anjos**, and J. Seixas, "Automatizing the Online Filter Test Management for a General-Purpose Particle Detector", *Computer Physics Communications*, 2009, doi: *10.1016/j.cpc.2010.10.003*.

2008 T. A. Collaboration, G. Aad, E. Abat, J. Abdallah, A. A. Abdelalim, A. Abdesselam, O. Abdinov, B. A. Abi, M. Abolins, H. Abramowicz, E. Acerbi, B. S. Acharya, R. Achenbach, M. Ackers, D. L. Adams, F. Adamyan, T. N. Addy, M. Aderholz, C. Adorisio, P. Adragna, M. Aharrouche, S. P. Ahlen, F. Ahles, A. Ahmad, H. Ahmed, G. Aielli, P. F. Åkesson, T. P. A. Åkesson, A. V. Akimov, S. M. Alam, J. Albert, S. Albrand, M. Aleksa, I. N. Aleksandrov, M. Aleppo, F. Alessandria, C. Alexa, G. Alexander, T. Alexopoulos, G. Alimonti, *et al*, "The ATLAS Experiment at the CERN Large Hadron Collider", *Journal of Instrumentation 3(08):S08003-S08003*, 2008, doi: *10.1088/1748-0221/3/08/s08003*.

2007 T. C. Xavier, **A. Anjos**, and J. M. de Seixas, "Discriminação Neural de Partículas para um Detector Submetido a uma Alta Taxa de Eventos", *Learning and Nonlinear Models - Revista da Sociedade Brasileira de Redes Neurais (SBRN) 4(2):79–92*, 2007, doi: *10.21528/lnlm-vol4-no2-art5*.

2006 **A. Anjos**, R. C. Torres, J. M. Seixas, B. C. Ferreira, and T. C. Xavier, "Neural triggering system operating on high resolution calorimetry information", *Nuclear Instruments and Methods in Physics Research 559:134–138*, 2006, doi: *10.1016/j.nima.2005.11.132*.

A. A. A. Collaboration, "Deployment of the ATLAS High-Level Trigger", *IEEE Transactions on Nuclear Science 53:2144–2149*, 2006, *https://api.zotero.org/users/5992358/publications/items/Q66Z8T7M/file*.

A. A. A. Collaboration, "A configuration system for the ATLAS trigger", *Journal of Instrumentation, Institute of Physics Publishing and Sissa(P05004)*, 2006, *https://api.zotero.org/users/5992358/publications/items/GZP5A97A/file*.

T. A. Collaboration, "The ATLAS Data Acquisition and Trigger : concept, design and status", *Nucl. Phys. B, Proc. Suppl. 172:178–182*, 2006, *https://api.zotero.org/users/5992358/publications/items/JUS44P7I/file*.

2005 T. A. Collaboration, "ATLAS DataFlow: the Read-Out Subsystem, Results from Trigger and Data-Acquisition System Testbed Studies and from Modeling", *IEEE Trans. Nucl. Sciences 53 (2006):912–917*, 2005, doi: *10.1109/rtc.2005.1547446*.

T. A. Collaboration, "Overview of the High-Level Trigger Electron and Photon Selection for the ATLAS Experiment at the LHC", *IEEE Transactions Nuclear Sciences (2005) 53:2839–2843*, 2005, *https://api.zotero.org/users/5992358/publications/items/XZ42WTX3/file*.

T. A. Collaboration, "Implementation and Performance of the Seeded Reconstruction for the ATLAS Event Filter Selection Software", *IEEE Trans. Nucl. Sciences 53 (2007):864–869*, 2005, *https://api.zotero.org/users/5992358/publications/items/6V2WJBVP/file*.

2004 J. T. Baines, C. P. Bee, A. Bogaerts, M. Bosman, D. Botterill, B. Caron, **A. Anjos**, F. Etienne, S. González, K. Karr, W. Li, C. Meessen, G. Merino, A. Negri, J. L. Pinfold, P. Pinto, Z. Qian, F. Touchard, P. Werner, S. Wheeler, F. J. Wickens, W. Wiedenmann, and G. Zobernig, "An Overview of the ATLAS High-Level Trigger Dataflow and Supervision", *IEEE Transaction on Nuclear Science 51(3):361–366*, 2004, doi: *10.1109/tns.2004.828875*.

T. A. Collaboration, "ATLAS TDAQ data collection software", *IEEE Transactions on Nuclear Science 51:585–590*, 2004, *https://api.zotero.org/users/5992358/publications/items/7WQKIXU3/file*.

T. A. Collaboration, "Studies for a common selection software environment in ATLAS : from the Level-2 Trigger to the offline reconstruction", *IEEE Transactions on Nuclear Science 51(3):915–920*, 2004, *https://api.zotero.org/users/5992358/publications/items/92TVWGMJ/file*.

T. A. Collaboration, "Architecture of the ATLAS high level trigger event selection software", *Nucl. Instrum. Methods Phys. Res. 518(1–2):537–541*, 2004, *https://api.zotero.org/users/5992358/publications/items/UZIF4P7D/file*.

T. A. Collaboration, "The base-line DataFlow system of the ATLAS Trigger and DAQ", *IEEE Transactions on Nuclear Science 51(3):470–475*, 2004, *https://api.zotero.org/users/5992358/publications/items/4893J5ZH/file*.

T. A. Collaboration, "Algorithms for the ATLAS high-level trigger", *IEEE Transactions on Nuclear Science 51(3):367–374*, 2004, *https://api.zotero.org/users/5992358/publications/items/G6S27R3P/file*.

A. A. A. Collaboration, "The Second Level Trigger of the ATLAS Experiment at CERN's LHC", *IEEE Transaction on Nuclear Science 51(3):909–914*, 2004, *https://api.zotero.org/users/5992358/publications/items/7AFJ5V4G/file*.

2003 **A. Anjos**, and J. Seixas, "Neural particle discrimination for triggering interesting physics channels with calorimetry data", *Nuclear Instruments And Methods In Physics Research A - Accelerators, Spectrometers, Detectors And Associated Equipament 502:713–715*, 2003, doi: *10.1016/s0168-9002(03)00553-9*.

T. A. Collaboration, "An Overview of Algorithms for the ATLAS High Level Trigger", *IEEE Transactions on Nuclear Science 51(3 (2004)):367–374*, 2003, *https://api.zotero.org/users/5992358/publications/items/746I7VIC/file*.

1998 J. Seixas, L. Caloba, **A. Anjos**, B. Kastrup, A. Dantas, and R. Linhares, "A neural online triggering system based on parallel processing", *IEEE Transactions on Nuclear Science 45(4):1814–1818*, 1998, doi: *10.1109/23.710942*.

Conference Papers

2026 T. van Rijn, R. Pulvirenti, M. Shiri, I. Meloni, O. Jimenez-del-Toro, M. Tomasoni, F. Hoogewoud, E. Kaisari, and **A. Anjos**, "Beyond the Last Frame: Temporal Modelling of Fluorescein Angiography for Hyperfluorescence Classification", in *Ophthalmic Medical Image Analysis (OMIA), MICCAI 2026 Workshops*, 2026, *https://api.zotero.org/users/5992358/publications/items/AC37PC75/file*.

G. Özbulak, O. Jimenez-del-Toro, L. Berton, and **A. Anjos**, "Loss-Conditioned Utility-Fairness Boundary Modeling for Medical Imaging", in *Fairness of AI in Medical Imaging (FAIMI), MICCAI 2026 Workshops*, 2026, *https://api.zotero.org/users/5992358/publications/items/25WSH8DP/file*.

M. Fatoretto, M. M. Raimundo, **A. Anjos**, and L. Berton, "A Dual-Constraint Optimization Approach for Algorithmic Fairness", in *2026 IEEE Congress on Evolutionary Computation (CEC):1-6*, 2026, doi: *10.1109/cec69861.2026.11727425*.

R. Pulvirenti, O. Jimenez-del-Toro, M. Tomasoni, F. Hoogewoud, and **A. Anjos**, "When Specialization Helps (and Hurts): Cross-Modality Transfer in Ophthalmic Imaging with Foundation Models", in *2026 IEEE 23rd International Symposium on Biomedical Imaging*, 2026, doi: *10.1109/isbi61048.2026.11515636*.

2025 D. Geissbuhler, A. Bornet, C. Marques, **A. Anjos**, S. Pereira, and D. Teodoro, "STM-GNN: Space-Time-and-Memory Graph Neural Networks for Predicting Multi-Drug Resistance Risks in Dynamic Patient Networks", in *International Conference on Artificial Intelligence in Medicine*, 2025, doi: *10.1007/978-3-031-95838-0_16*.

M. Vitek, D. Tomašević, A. Das, S. Nathan, G. Özbulak, G. A. T. Özbulak, J. Calbimonte, **A. Anjos**, H. H. Bhatt, D. D. Premani, J. Chaudhari, C. Wang, J. Jiang, C. Zhang, Q. Zhang, I. I. Ganapathi, S. S. Ali, D. Velayudan, M. Assefa, N. Werghi, Z. A. Daniels, L. John, R. Vyas, J. N. Khiarak, T. A. Saeed, M. Nasehi, A. Kianfar, M. P. Panahi, G. Sharma, P. R. Panth, R. Ramachandra, A. Nigam, U. Pal, H. Pedrini, and V. Struc, "Privacy-enhancing Sclera Segmentation Benchmarking Competition: SSBC 2025", in *2025 IEEE International Joint Conference on Biometrics (IJCB):1-13*, 2025, doi: *10.1109/ijcb65343.2025.11411539*.

2024 Ö. A. Güler, M. Günther, and **A. Anjos**, "Refining Tuberculosis Detection in CXR Imaging: Addressing Bias in Deep Neural Networks via Interpretability", in *2024 12th European Workshop on Visual Information Processing (EUVIP):1-6*, 2024, doi: *10.1109/euvip61797.2024.10772829*.

D. Queiroz, **A. Anjos**, and L. Berton, "Using Backbone Foundation Model for Evaluating Fairness in Chest Radiography Without Demographic Data", in *Ethics and Fairness in Medical Imaging:109-118*, 2024, doi: *10.1007/978-3-031-72787-0_11*.

D. Q. Neto, A. Carlos, M. Fatoretto, L. F. Nakayama, **A. Anjos**, and L. Berton, "Does Data-Efficient Generalization Exacerbate Bias in Foundation Models?", in *Fairness and Ethics Towards Transparent AI: Facing the Challenge through Model Debiasing (FAILED), ECCV 2024 Workshops*, 2024, doi: *10.48550/arxiv.2408.16154*.

O. Jimenez-del-Toro, C. Aberle, R. Schaer, M. Bach, K. Flouris, E. Konukoglu, B. Stieltjes, M. M. Obmann, **A. Anjos**, H. Müller, and A. Depeursinge, "Comparing Stability and Discriminatory Power of Hand-crafted Versus Deep Radiomics:

A 3D-Printed Anthropomorphic Phantom Study", in *2024 12th European Workshop on Visual Information Processing (EUVIP)*, 2024, doi: *10.1109/euvip61797.2024.10772813*.

2023 V. Amiot, O. Jimenez-Del-Toro, P. Eyraud, Y. Guex-Crosier, C. Bergin, **A. Anjos**, F. Hoogewoud, and M. Tomasoni, "Fully Automatic Grading of Retinal Vasculitis on Fluorescein Angiography Time-lapse from Rheumatologic Uveitis", in *2023 IEEE 36th International Symposium on Computer-Based Medical Systems (CBMS)*, 2023, doi: *10.1109/cbms58004.2023.00301*.

2021 M. A. Renzo, N. Fernandez, A. A. Baceti, N. N. M. Junior, and **A. Anjos**, "Development of a lung segmentation algorithm for analog imaged chest X-Ray: preliminary results", in *Anais do 15. Congresso Brasileiro de Inteligência Computacional:1-8*, 2021, doi: *10.21528/cbic2021-123*.

2018 O. Nikisins, T. Eglitis, **A. Anjos**, and S. Marcel, "Fast cross-correlation based wrist vein recognition algorithm with rotation and translation compensation", in *Sixth International Workshop on Biometrics and Forensics*, 2018, doi: *10.1109/iwbf.2018.8401550*.

O. Nikisins, A. Mohammadi, **A. Anjos**, and S. Marcel, "On Effectiveness of Anomaly Detection Approaches against Unseen Presentation Attacks in Face Anti-Spoofing", in *The 11th IAPR International Conference on Biometrics (ICB 2018)*, 2018, doi: *10.1109/icb2018.2018.00022*.

2017 M. Cernak, A. Komaty, A. Mohammadi, **A. Anjos**, and S. Marcel, "Bob Speaks Kaldi", in *Proceedings of Interspeech*, 2017, *https://api.zotero.org/users/5992358/publications/items/5RRHGBTA/file*.

A. Anjos, M. Günther, T. F. Pereira, P. Korshunov, A. Mohammadi, and S. Marcel, "Continuously Reproducing Toolchains in Pattern Recognition and Machine Learning Experiments", in *Thirty-fourth International Conference on Machine Learning*, 2017, *https://api.zotero.org/users/5992358/publications/items/R55HB7JQ/file*.

A. Anjos, L. E. Shafey, and S. Marcel, "BEAT: An Open-Science Web Platform", in *Thirty-fourth International Conference on Machine Learning*, 2017, *https://api.zotero.org/users/5992358/publications/items/ZQ5WW5UB/file*.

2013 I. Chingovska, **A. Anjos**, and S. Marcel, "Anti-spoofing in action: joint operation with a verification system", in *Computer Vision and Pattern Recognition Conference - Biometrics Workshop*, 2013, doi: *10.1109/cvprw.2013.22*.

J. Komulainen, A. Hadid, M. Pietikäinen, **A. Anjos**, and S. Marcel, "Complementary Countermeasures for Detecting Scenic Face Spoofing Attacks", in *International Conference on Biometrics 2013*, 2013, doi: *10.1109/icb.2013.6612968*.

T. F. Pereira, **A. Anjos**, J. M. De Martino, and S. Marcel, "Can face anti-spoofing countermeasures work in a real world scenario?", in *International Conference on Biometrics 2013*, 2013, doi: *10.1109/icb.2013.6612981*.

I. Chingovska, J. Yang, Z. Lei, D. Yi, S. Z. Li, O. Kähm, C. Glaser, N. Damer, A. Kuijper, A. Nouak, J. Komulainen, T. Pereira, S. Gupta, S. Khandelwal, S. Bansal, A. Rai, T. Krishna, D. Goyal, M. Waris, H. Zhang, I. Ahmad, S. Kiranyaz, M. Gabbouj, R. Tronci, M. Pili, N. Sirena, F. Roli, J. Galbally, J. Fierrez, A. Pinto, H. Pedrini, W. S. Schwartz, A. Rocha, **A. Anjos**, and S. Marcel, "The 2nd Competition on Counter Measures to 2D Face Spoofing Attacks", in *International Conference on Biometrics 2013*, 2013, doi: *10.1109/icb.2013.6613026*.

2012 I. Chingovska, **A. Anjos**, and S. Marcel, "On the Effectiveness of Local Binary Patterns in Face Anti-spoofing", in *IEEE International Conference of the Biometrics Special Interest Group*, 2012, *https://api.zotero.org/users/5992358/publications/items/HCXQES2H/file*.

A. Anjos, L. E. Shafey, R. Wallace, M. Günther, C. McCool, and S. Marcel, "Bob: a free signal processing and machine learning toolbox for researchers", in *ACM Multimedia 2012:1449–1452*, 2012, doi: *10.1145/2393347.2396517*.

T. F. Pereira, **A. Anjos**, J. M. De Martino, and S. Marcel, "LBP-TOP based countermeasure against facial spoofing attacks", in *International Workshop on Computer Vision With Local Binary Pattern Variants*, 2012, doi: *10.1007/978-3-642-37410-4_11*.

2011 **A. Anjos**, and S. Marcel, "Counter-Measures to Photo Attacks in Face Recognition: a public database and a baseline", in *International Joint Conference on Biometrics 2011*, 2011, doi: *10.1109/ijcb.2011.6117503*.

M. M. Chakka, **A. Anjos**, S. Marcel, and others, "Competition on Counter Measures to 2-D Facial Spoofing Attacks", in *International Joint Conference on Biometrics 2011*, 2011, doi: *10.1109/ijcb.2011.6117509*.

2009 T. A. Collaboration, "ATLAS Trigger and Data Acquisition: capabilities and commissioning", in *11th Pisa Meeting on Advanced Detectors 617:306–309*, 2009, doi: *10.1016/j.nima.2009.06.114*.

T. A. Collaboration, "Configuration and Control of the ATLAS Trigger and Data Acquisition", in *The 1st international conference on Technology and Instrumentation in Particle Physics*, 2009, *https://api.zotero.org/users/5992358/publications/items/35UTZG93/file*.

T. A. Collaboration, "Atlas trigger for first physics and beyond", in *Physics at LHC 2008 29 September - October 4, 2008 Split, Croatia*, 2009, *https://api.zotero.org/users/5992358/publications/items/2PZ994WK/file*.

T. A. Collaboration, "ATLAS Trigger Status and Results From Commissioning Operations", in *Advanced Computing on High-Energy Physics 2008, Erice, Sicily, Italy*, 2009, *https://api.zotero.org/users/5992358/publications/items/DFDAJJJ4/file*.

T. A. Collaboration, "Commissioning of the ATLAS High Level Trigger with Single Beam and Cosmic Rays", in *Computing in High Energy and Nuclear Physics, Prague, Czech Republic, 21 - 27 Mar 2009*, 2009, *https://api.zotero.org/users/5992358/publications/items/5FI6WSB5/file*.

T. A. Collaboration, "The ATLAS online High Level Trigger framework: experience reusing offline software components in the ATLAS trigger", in *Computing in High Energy and Nuclear Physics, Prague, Czech Republic, 21 - 27 Mar 2009*, 2009, *https://api.zotero.org/users/5992358/publications/items/HBJRN7GR/file*.

2008 **A. Anjos**, "Trigger Systems", in *Experimental High-Energy Physics and Associated Technologies Workshop*, 2008.

A. A. A. Collaboration, "The DAQ/HLT system of the ATLAS experiment", in *International Workshop on Advanced Computing and Analysis Techniques in Physics Research*, 2008, *https://api.zotero.org/users/5992358/publications/items/SQENSHCA/file*.

T. A. Collaboration, "Readiness of the ATLAS Trigger and Data Acquisition system for the first LHC beams", in *11th Topical Seminar On Innovative Particle And Radiation Detectors, Siena, Italy*, 2008, *https://api.zotero.org/users/5992358/publications/items/DB5TJGDP/file*.

2007 T. A. Collaboration, "Alignment data streams for the ATLAS Inner Detector", in *Computing for High-Energy Physics*, 2007, doi: *10.1088/1742-6596/119/2/022030*.

A. A. A. Collaboration, "The Configuration System of the ATLAS Trigger", in *IEEE Real-time conference*, 2007, *https://api.zotero.org/users/5992358/publications/items/5X23ZBK7/file*.

T. A. Collaboration, "Performance of the final Event Builder for the ATLAS Experiment", in *15th IEEE Real Time Conference 2007*, 2007, *https://api.zotero.org/users/5992358/publications/items/K3M5MUZG/file*.

T. A. Collaboration, "Integration of the Trigger and Data Acquisition systems in ATLAS", in *IEEE Real-time conference*, 2007, *https://api.zotero.org/users/5992358/publications/items/8E888TIM/file*.

T. A. Collaboration, "The ATLAS trigger - high-level trigger commissioning and operation during early data taking", in *International Europhysics Conference on High Energy Physics*, 2007, *https://api.zotero.org/users/5992358/publications/items/KRZ7KPS2/file*.

T. A. Collaboration, "The ATLAS Trigger - Commissioning with cosmic rays", in *International Conference on Computing in High Energy and Nuclear Physics*, 2007, *https://api.zotero.org/users/5992358/publications/items/UC7BVECC/file*.

R. C. Torres, J. M. Seixas, **A. Anjos**, and D. V. Cunha, "Online Electron/Jet Neural High-Level Trigger over Indenpendent Calorimetry Information", in *XI International Workshop on Advanced Computing and Analysis Techniques in Physics Research*, 2007, *https://api.zotero.org/users/5992358/publications/items/2VHMDEJM/file*.

T. A. Collaboration, "The ATLAS Event Builder", in *IEEE Nuclear Science Symposium and Medical Imaging Conference*, 2007, *https://api.zotero.org/users/5992358/publications/items/PWF3I3E7/file*.

2006 T. A. Collaboration, "Testing on a Large Scale: running the ATLAS Data Acquisition and High Level Trigger Software on 700 PC Nodes", in *Computing In High Energy and Nuclear Physics*, 2006, *https://api.zotero.org/users/5992358/publications/items/753XBFJ3/file*.

T. A. Collaboration, "Studies with the ATLAS Trigger and Data Acquisition Pre-Series Setup", in *15th International Conference on Computing In High Energy and Nuclear Physics*, 2006, *https://api.zotero.org/users/5992358/publications/items/SWU3XHMQ/file*.

T. A. Collaboration, "ATLAS High Level Trigger Infrastructure, ROI Collection and Event Building", in *15th International Conference on Computing In High Energy and Nuclear Physics*, 2006, *https://api.zotero.org/users/5992358/publications/items/Q5FXBHXA/file*.

2005 A. A. A. Collaboration, "Configuration of the ATLAS trigger", in *14th IEEE NPSS Real Time Conference:990–994*, 2005, *https://api.zotero.org/users/5992358/publications/items/4AIDEPTG/file*.

A. Anjos, R. C. Torres, B. C. Ferreira, T. C. Xavier, J. M. Seixas, and D. O. Damazio, "Otimização do Sistema de Trigger do Segundo Nível do ATLAS Baseado em Calorimetria", in *XXVI Encontro Nacional de Física de Partículas e Campos*, 2005, *https://api.zotero.org/users/5992358/publications/items/FIZVWDGQ/file*.

T. A. Collaboration, "Muon Reconstruction and Identification for the Event Filter of the ATLAS experiment", in *9th ICATAPP Conference on High Energy Physics*, 2005, *https://api.zotero.org/users/5992358/publications/items/7W4K2GCW/file*.

T. A. Collaboration, "Implementation and performance of a tau lepton selection within the ATLAS trigger system at the LHC", in *9th ICATPP Conference on Astroparticle, Particle, Space Physics, Detectors and Medical Physics Applications*, 2005, *https://api.zotero.org/users/5992358/publications/items/BVRS98CP/file*.

A. Anjos, R. C. Torres, B. C. Ferreira, T. C. Xavier, and J. M. de Seixas, "Discriminação Neural de Elétrons no Segundo Nível de Trigger do ATLAS", in *XXVI Encontro Nacional de Física de Partículas e Campos*, 2005, *https://api.zotero.org/users/5992358/publications/items/KQW3EJ5W/file*.

2004 **A. Anjos**, and J. M. Seixas, "Os Filtros de Alto Nível do Experimento ATLAS", in *XXVI Encontro Nacional de Física de Partículas e Campos*, 2004, *https://api.zotero.org/users/5992358/publications/items/EZT4G27A/file*.

T. A. Collaboration, "Performance of the ATLAS DAQ DataFlow system", in *Computing in High Energy Physics and Nuclear Physics*, 2004, doi: *10.5170/cern-2005-002.91*.

T. A. Collaboration, "Online Muon Reconstruction in the ATLAS Level-2 trigger system", in *Nuclear Science Symposium and Medical Imaging Conference*, 2004, *https://api.zotero.org/users/5992358/publications/items/VBVZT92N/file*.

T. A. Collaboration, "Portable Gathering System for Monitoring and Online Calibration at ATLAS", in *Computing in High Energy Physics and Nuclear Physics 2004*, 2004, *https://api.zotero.org/users/5992358/publications/items/76ICCDF8/file*.

T. A. Collaboration, "Implementation and Performance of the High Level Trigger Electron and Photon Selection for the ATLAS Experiment at the LHC", in *IEEE Nuclear Science Symposium and Medical Imaging Conference*, 2004, *https://api.zotero.org/users/5992358/publications/items/V74FGXF7/file*.

T. A. Collaboration, "Design, deployment and functional tests of the on-line Event Filter for the ATLAS experiment at LHC", in *Nuclear Science Symposium and Medical Imaging Conference*, 2004, *https://api.zotero.org/users/5992358/publications/items/FZHVD6CE/file*.

2003 T. A. Collaboration, "Experience with multi-threaded C++ applications in the ATLAS dataflow software", in *Conference for Computing in High-Energy and Nuclear Physics*, 2003, *https://api.zotero.org/users/5992358/publications/items/C5H5GAMD/file*.

T. A. Collaboration, "A New Implementation of the Region-of-Interest Strategy for the ATLAS Second Level Trigger", in *Conference for Computing in High-Energy and Nuclear Physics*, 2003, *https://api.zotero.org/users/5992358/publications/items/2EFDCRZJ/file*.

T. A. Collaboration, "The Algorithm Steering and Trigger Decision mechanism of the ATLAS High Level Trigger", in *Conference for Computing in High-Energy and Nuclear Physics*, 2003, *https://api.zotero.org/users/5992358/publications/items/GAGP9ESQ/file*.

T. A. Collaboration, "Architecture of the ATLAS online physics-selection software at LHC", in *Conference on Astroparticle, Particle, Space Physics, Detectors and Medical Physics Applications*, 2003, *https://api.zotero.org/users/5992358/publications/items/QW5QZ3G5/file*.

T. A. Collaboration, "The baseline dataflow system of the ATLAS trigger and DAQ", in *9th Workshop on Electronics for LHC Experiments*, 2003, *https://api.zotero.org/users/5992358/publications/items/J3C6B9JF/file*.

T. A. Collaboration, "The DataFlow System of the ATLAS Trigger and DAQ", in *Conference for Computing in High-Energy and Nuclear Physics*, 2003, *https://api.zotero.org/users/5992358/publications/items/ETM97WMJ/file*.

2001 **A. Anjos**, and J. Seixas, "Redes Neurais especialistas para a separação Elétron-Jato usando Calorímetros multi-camadas e multi-segmentados", in *XXII Encontro Nacional de Física de Partículas e Campos*, 2001, *https://api.zotero.org/users/5992358/publications/items/47S2VMRK/file*.

2000 **A. Anjos**, and J. M. de Seixas, "Mapeamento em anéis para uma separação neuronal elétron-jato usando calorímetros multi-camadas e multi-segmentados", in *XIX Encontro Nacional de Física de Partículas e Campos*, 2000, *https://api.zotero.org/users/5992358/publications/items/PIQHGV99/file*.

1999 **A. Anjos**, and J. M. de Seixas, "Integrando Plataformas e Algoritmos para o Segundo Nível de Trigger do Experimento ATLAS", in *Encontro Nacional de Física de Partículas e Campos*, 1999, *https://api.zotero.org/users/5992358/publications/items/ADE5P7D9/file*.

1998 **A. Anjos**, A. Dantas, and J. M. de Seixas, "Um Protótipo do Sistema de Validação do Nível 2 para as Condições do LHC", in *Encontro Nacional de Física de Partículas e Campos:32–33*, 1998, *https://api.zotero.org/users/5992358/publications/items/AE3ZBAWP/file*.

J. M. Seixas, **A. Anjos**, C. B. Prado, L. P. Calôba, A. C. H. Dantas, and J. C. R. Aguiar, "Neural classifiers implemented in a transputer based parallel machine", in *International Meeting on Vector and Parallel Processing (VECPAR)*, 1998, *https://api.zotero.org/users/5992358/publications/items/9VMFH959/file*.

1997 J. Seixas, L. Calôba, **A. Anjos**, A. Dantas, and R. Linhares, "Fast Neural Decision System Based On DSPs And Parallel Processing", in *International Conference on Signal Processing Applications and Technologies, San Diego, USA:1629–1633*, 1997, *https://api.zotero.org/users/5992358/publications/items/XMCRWBKF/file*.

1996 J. Seixas, L. Calôba, and **A. Anjos**, "Particle discrimination using sub-optimal filtering techniques", in *Congresso Brasileiro de Automatica (CBA), São Paulo, Brasil:635–640*, 1996, *https://api.zotero.org/users/5992358/publications/items/KMMH9868/file*.

Book Chapters

2019 I. Chingovska, A. Mohammadi, **A. Anjos**, and S. Marcel, "Evaluation Methodologies for Biometric Presentation Attack Detection", *Handbook of Biometric Anti-Spoofing:457–480*, 2019, doi: *10.1007/978-3-319-92627-8_20*.

A. Anjos, P. Tome, and S. Marcel, "An Introduction to Vein Presentation Attacks and Detection", *Handbook of Biometric Anti-Spoofing:419–438*, 2019, doi: *10.1007/978-3-319-92627-8_18*.

S. Bhattacharjee, A. Mohammadi, **A. Anjos**, and S. Marcel, "Recent Advances in Face Presentation Attack Detection", *Handbook of Biometric Anti-Spoofing:207–228*, 2019, doi: *10.1007/978-3-319-92627-8_10*.

2016 I. Chingovska, N. Erdogmus, **A. Anjos**, and S. Marcel, "Face Recognition Systems Under Spoofing Attacks", *Face Recognition Systems Under Spoofing Attacks:165–194*, 2016, doi: *10.1007/978-3-319-28501-6_8*.

2014 S. Z.Li, J. Galbally, **A. Anjos**, and S. Marcel, "Evaluation Databases", *Handbook of Biometric Anti-Spoofing:247–278*, 2014, doi: *10.1007/978-1-4471-6524-8*.

A. Anjos, J. Komulainen, S. Marcel, A. Hadid, and M. Pietikainen, "Face Anti-Spoofing: Visual Approach", *Handbook of Biometric Anti-Spoofing:65–82*, 2014, doi: *10.1007/978-1-4471-6524-8_4*.

I. Chingovska, **A. Anjos**, and S. Marcel, "Evaluation Methodologies", *Handbook of Biometric Anti-Spoofing:185–204*, 2014, doi: *10.1007/978-1-4471-6524-8_10*.

I. Chingovska, **A. Anjos**, and S. Marcel, "Anti-spoofing: Evaluation Methodologies", *Encyclopedia of Biometrics*, 2014, doi: *10.1007/978-3-642-27733-7*.

A. Anjos, I. Chingovska, and S. Marcel, "Anti-Spoofing: Face Databases", *Encyclopedia of Biometrics*, 2014, doi: *10.1007/978-3-642-27733-7_9212-2*.

Preprints

2025 D. Geissbuhler, A. Bornet, C. Marques, **A. Anjos**, S. Pereira, and D. Teodoro, "STM-GNN: Space-Time-and-Memory Graph Neural Networks for Predicting Multi-Drug Resistance Risks in Dynamic Patient Networks", *Health Informatics*, 2025, doi: *10.1101/2025.05.27.25327491*.

2024 V. Amiot, O. Jimenez-del-Toro, Y. Guex-Croisier, M. Ott, T. Bogaciu, S. Banerjee, J. Howell, C. Amstutz, C. Chiquet, C. Bergin, I. Meloni, M. Tomasoni, F. Hoogewoud, and **A. Anjos**, "Automatic Transformer-Based Grading of Multiple Retinal Inflammatory Signs on Fluorescein Angiography", *SSRN*, 2024, doi: *10.2139/ssrn.4960069*.

2020 A. Galdran, **A. Anjos**, J. Dolz, H. Chakor, H. Lombaert, and I. B. Ayed, "The Little W-Net That Could: State-of-the-Art Retinal Vessel Segmentation with Minimalistic Models", *arXiv*, 2020, doi: *10.48550/arxiv.2009.01907*.

T. Laibacher, and **A. Anjos**, "On the Evaluation and Real-World Usage Scenarios of Deep Vessel Segmentation for Retinography", *arXiv*, 2020, doi: *10.48550/arxiv.1909.03856*.

2017 **A. Anjos**, L. El-Shafey, and S. Marcel, "BEAT: An Open-Source Web-Based Open-Science Platform", *arXiv*, 2017, doi: *10.48550/arxiv.1704.02319*.

G. Heusch, **A. Anjos**, and S. Marcel, “A reproducible study on remote heart rate measurement”, *arXiv*, 2017, doi: *10.48550/arxiv.1709.00962*.

Technical Reports

2008 T. A. Collaboration, “Expected Performance of the ATLAS Experiment Detector, Trigger, Physics”, *CERN:arXiv:0901.0512*, 2008, doi: *10.48550/arxiv.0901.0512*.

2003 T. A. Collaboration, “The ATLAS HLT, DAQ and DCS Technical Design Report”, *CERN Publication*, 2003, *https://api.zotero.org/users/5992358/publications/items/53DHC686/file*.

Patents

2019 L. S. Momo, L. C. Torres, S. Marcel, **A. Anjos**, M. Liebling, A. Shajkofci, S. Amoos, A. Woeffray, A. Sierro, P. Roduit, P. Ferrez, and L. Bonvin, “Method and Device for Biometric Vascular Recognition and/or Identification”,*:WO/2019/150254*, 2019, *https://api.zotero.org/users/5992358/publications/items/QG2BP7FM/file*.

2018 S. Marcel, **A. Anjos**, and P. Abbet, “Method and internet-connected server for reviewing a computer-executable experiment”,*:US9973503B2*, 2018, *https://api.zotero.org/users/5992358/publications/items/A2JXK46I/file*.

2017 **A. Anjos**, and S. Marcel, “A data-network connected server, a device, a platform and a method for conducting computer-executable experiments”,*:WO/2017/221049*, 2017, *https://api.zotero.org/users/5992358/publications/items/93HG875F/file*.

Theses

2006 **A. Anjos**, “Sistema Online de Filtragem em um Ambiente com Alta Taxa de Eventos”,, 2006, *https://api.zotero.org/users/5992358/publications/items/6SWJXRDZ/file*.

2001 **A. Anjos**, “Sistema neuronal rápido de decisão baseado em calorimetria de altas energias”,, 2001, *https://api.zotero.org/users/5992358/publications/items/C8FM5BTA/file*.

1997 **A. Anjos**, “Sistema de classificação baseado em uma máquina com sistema distribuído”,, 1997, *https://api.zotero.org/users/5992358/publications/items/3MENIXQM/file*.